

Debate-Enhanced Fact Verification Adversarial Argumentative Reasoning for Claim Assessment

Shir Babian Yael Batat Gilad Ticher

Hebrew University of Jerusalem

{shir.babian, yael.batat, gilad.ticher}@mail.huji.ac.il

Abstract

Most automated fact verification systems use direct classification approaches that may miss the nuanced reasoning inherent in human fact checking. We investigate whether simulated debates between opposing agents improve fact verification accuracy. We fine-tune FLAN-T5-large (Chung et al., 2022) on Reddit’s r/ChangeMyView dataset (Tan et al., 2016) for argumentative reasoning, then simulate debates between agents supporting and refuting claims using FEVER benchmark evidence (Thorne et al., 2018). Debate transcripts provide context for final verification decisions. Evaluation on FEVER’s three-class labels (SUPPORTS, REFUTES, NOT ENOUGH INFO) compares debate enhanced predictions against baseline methods. Results show modest improvements for the fine-tuned model when using short debate sequences, suggesting that structured argumentative reasoning can enhance automated fact verification under specific conditions.

1 Introduction

The proliferation of misinformation across digital platforms has emerged as a critical challenge, with user generated content growth outpacing manual fact-checking capabilities and necessitating automated verification systems (Kotonya and Toni, 2020).

Current automated fact verification systems predominantly employ single step neural classification, directly predicting claim veracity from retrieved evidence (Thorne et al., 2018). However, these approaches may not capture the complex reasoning inherent in human fact checking, which involves deliberative evaluation of multiple perspectives and iterative refinement through discourse (Popat et al., 2017).

Recent advances in multi-agent debate based reasoning have shown promise for complex reasoning tasks by leveraging adversarial interactions

to expose weaknesses and generate robust conclusions (Du et al., 2023). Despite the inherently argumentative nature of claim evaluation, the application of debate based methods to fact verification remains underexplored.

This work investigates whether simulated debates between opposing LLM agents can improve automated fact verification accuracy. We fine-tune FLAN-T5-large on argumentative discourse from Reddit’s r/ChangeMyView dataset (Tan et al., 2016; Chung et al., 2022), then deploy these agents to debate claims using FEVER benchmark evidence. Our experimental evaluation quantifies the impact of debate enhanced reasoning on fact verification performance, contributing to understanding how multi-agent argumentative frameworks can advance automated claim assessment.¹

2 Data

2.1 r/ChangeMyView (CMV) Dataset

We utilize 10 structured argumentative discussions from Reddit’s r/ChangeMyView subreddit, where users post claims beginning with "CMV:" and engage in persuasive discourse. Stance annotations (support/refute/no_stance) for each comment are automatically generated using the all-MiniLM-L6-v2 sentence transformer model (Reimers and Gurevych, 2019).

The dataset includes metadata for successful persuasion attempts identified through DeltaBot confirmations, where original posters award "delta" points for perspective-changing comments. We extract debate training pairs using arguments from delta-awarded users as high-quality evidence for persuasive reasoning.

From each conversation tree, we create prompt-response pairs including the claim, evidence from

¹ <https://github.com/gilas19/DEFV>

delta responses, debate role, and context (see Appendix A.1.2). Responses are limited to 3 sentences and 200 characters, with filtering applied to remove short responses (<50 characters), system messages, and deleted content.

2.2 FEVER Dataset

We use the FEVER Gold Evidence dataset (Atanasova et al., 2020) from HuggingFace, containing claims with pre-extracted Wikipedia evidence sentences. Claims are annotated as *SUPPORTS*, *REFUTES*, or *NOT ENOUGH INFO*.

Evidence text is extracted from [title, line_number, text] tuples, concatenated with spaces, and truncated to 500 characters for computational efficiency. We evaluate on 120 claims while preserving uniform label distribution, with predictions standardized through post-processing to match valid labels.

3 Methods

3.1 Model Architecture and Fine-tuning

We use the google/flan-t5-large model as the base architecture for all experiments (see Figure 1 for the complete system architecture). FLAN-T5 is a sequence-to-sequence language model pre-trained with instruction tuning, making it suitable for multistep reasoning tasks such as simulated debates. To enhance its ability to produce high quality argumentative responses, we fine-tuned the model on CMV discussion data using Low-Rank Adaptation (LoRA) (Hu et al., 2021). System Architecture can

We applied LoRA adapters to the self-attention layers of the FLAN-T5 model, reducing the number of trainable parameters. The LoRA configuration used rank $r = 16$, $\alpha = 32$, and dropout rate of 0.1. Fine-tuning was performed for 2 epochs with a batch size of 2, a learning rate of $4e - 5$, and early stopping based on validation loss. Initial experiments with 5 epochs showed significant overfitting (see in Appendix A.3), leading us to reduce the training duration. The maximum input and target sequence lengths were set to 512 and 256 tokens, respectively. The training was conducted using an NVIDIA T4 GPU with precision BF16.

3.2 Prompting and Debate Simulation

To evaluate the system, we compare two settings: (1) a direct fact verification prompt using claim

and evidence only (no-debate), and (2) a simulated debate between two agents who argue opposing positions before a final prediction is made (debate based). In the debate based setting, the agents alternate turns (1, 2, 4, 6 each) and are prompted explicitly to support or refute the claim using the evidence, depending on their assigned role. Prompts include instructions enforcing role consistency, dialog history, and reminders to maintain argumentative stance (see Appendix A.1.2).

After simulating the debate, a final classification prompt summarizes the dialogue and asks the model to output one of three FEVER labels.(see Appendix A.1.3)

3.3 Evaluation Setup

Evaluation was conducted over a set of configurations, varying the number of debate turns and the initial speaker (support or refute). For each configuration, we measure accuracy, average generation time, and token length. Final results are saved and analyzed using scripts that compute detailed error distributions, confusion matrices, and correlation statistics across model types and debate strategies.

4 Results

4.1 Overall Accuracy

Table 1 presents the accuracy scores across all experimental configurations. Both the baseline and fine-tuned models perform comparably, with accuracies ranging from 57.5% to 66.7%. The no-debate baseline model achieved 60.8% accuracy, while the fine-tuned model under the same condition reached 57.5%. Among debate configurations, increasing the number of turns (1 to 6) or changing the initiating agent (Agent 1 vs. Agent 2) did not result in a consistent improvement. The highest accuracy (66.7%) was observed in the baseline model with 4 turns initiated by Agent 1, but other configurations remained close in performance.

Performance was strongly influenced by prompt phrasing and structure, particularly in the early stages of development. Prompt refinements led to improved stance consistency, argument relevance, and overall accuracy (see Appendix A.2). Additionally, longer fine-tuning schedules appeared to induce overfitting to conversational patterns in the training data, which was mitigated by reducing to 2 epochs (see Appendix A.3).

4.2 Label Prediction Patterns

Figures 2, 3 and 4 show the prediction distributions by gold label and number of debate turns for the baseline and fine-tuned models, respectively. For claims labeled *SUPPORTS*, both models performed with high consistency, predicting the correct label in nearly all cases regardless of debate length. For *REFUTES*, predictions were more varied, though still reasonably accurate.

The most notable discrepancies were observed for the *NOT ENOUGH INFO* class. The baseline model (Figure 3) frequently misclassified these claims, most often labeling them as *SUPPORTS*, followed by *REFUTES*, with the correct label appearing least. In contrast, the fine-tuned model (Figure 4) also showed a preference for *SUPPORTS*, but exhibited a slightly better distribution by assigning the correct *NOT ENOUGH INFO* label more often.

Similar patterns are observed in the no-debate condition. The same pattern persists: the fine-tuned model predicts *NOT ENOUGH INFO* more frequently and accurately than the baseline, but both models still exhibit a strong bias toward *SUPPORTS* when handling uncertain claims.

5 Analysis

The results deviate from our initial hypotheses. We expected that the fine-tuned model, having been trained on structured debate-style interactions, would outperform the baseline especially in multi-turn dialogue settings. Similarly, we anticipated that longer debates (e.g., 4 or 6 turns) would allow agents to exchange more nuanced arguments, thereby improving final judgment accuracy. However, our results show that the accuracy remains largely stable between models and configurations, with no consistent gains from fine-tuning or increased depth of debate (Table 1).

A plausible explanation lies in the model architecture. FLAN-T5-large, while powerful, may struggle with long, structured prompts, particularly when the prompt includes extended debate history and evidence. As prompt length increases, earlier content may be lost due to truncation or lack of attention, reducing the benefit of extended dialogue. Furthermore, the final decision prompt may implicitly prioritize the evidence and claim, leading the model to ignore the intervening debate content, even when it exists.

The error analysis reinforces these findings.

Both models consistently perform well on clearly labeled *SUPPORTS* and *REFUTES* claims, but show substantial confusion on *NOT ENOUGH INFO*. As seen in Figure 2, the baseline model frequently overpredicts *SUPPORTS* or *REFUTES*, likely due to overconfidence in partial or misleading evidence. The fine-tuned model (Figure 4) improves marginally, assigning the correct *NOT ENOUGH INFO* label more frequently, suggesting a better uncertainty calibration. However, the improvement is limited, indicating that fine-tuning alone may not be sufficient to resolve ambiguity in poorly evidenced claims.

6 Conclusion

We investigated whether simulated debates between opposing agents can enhance automated fact verification using FLAN-T5-large fine-tuned on r/ChangeMyView data. The Results showed only modest accuracy improvements with no consistent gains from increased debate turns or different initiating strategies. Both baseline and fine-tuned models achieved similar performance, suggesting current model architecture may not effectively leverage extended argumentative context. While structured debate holds theoretical promise, realizing substantial improvements requires advances in model architecture and training methodologies.

7 Safety and Limitations

Fine-tuning on Reddit data may introduce social biases, and the system’s ability to generate opposing arguments could be misused for misleading content. Persistent misclassification of *NOT ENOUGH INFO* cases may lead to overconfident predictions. Our evaluation is limited by a small dataset (120 claims), reliance on Reddit discourse patterns and English-only scope. Additionally, the debate process generates variable conversation patterns that can lead to inconsistent classification decisions, reducing prediction reliability compared to deterministic approaches.

Beginning Stance	Model Type	Number of Turns				
		w/o debate	1	2	4	6
Agent 1 (Support)	Baseline	0.61	0.64	0.63	0.67	0.62
	Fine-tuned	0.58	0.65	0.59	0.65	0.64
Agent 2 (Refute)	Baseline	0.61	0.59	0.66	0.63	0.63
	Fine-tuned	0.58	0.62	0.62	0.62	0.61
Average	Baseline	0.61	0.62	0.64	0.65	0.63
	Fine-tuned	0.58	0.63	0.60	0.63	0.62

Table 1: Performance comparison of baseline and fine-tuned models across different debate configurations. Results show accuracy values for varying numbers of debate turns, with separate evaluation for support and refute initiating strategies.

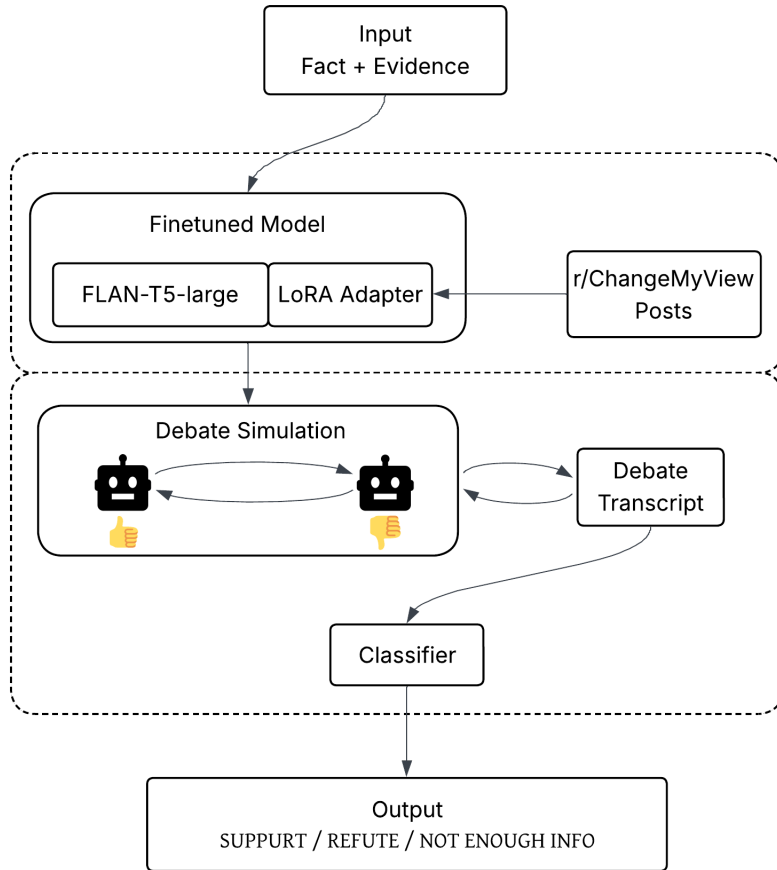


Figure 1: Architecture of the debate-enhanced fact verification system. The system takes a factual claim and evidence as input, processes them through a FLAN-T5-large model fine-tuned with LoRA adapters on r/ChangeMyView data. Two agents with opposing stances (support and refute) engage in simulated debate, generating a transcript that provides enriched context for the final three-class classification decision.

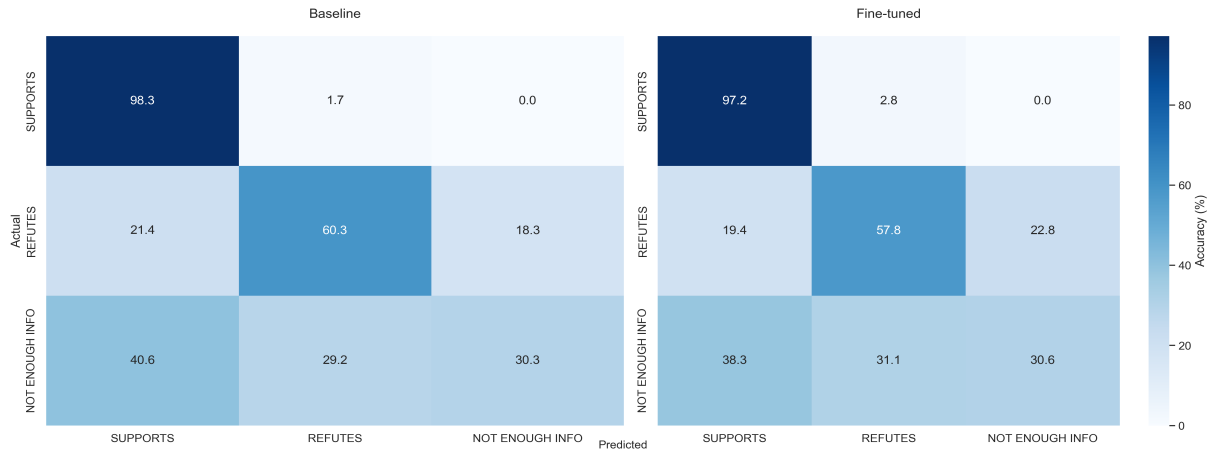


Figure 2: Confusion matrices showing 'true' versus 'predicted' labels for the baseline model (left) and fine-tuned model (right) across all debate configurations. Values represent percentages of predictions for each true label category.

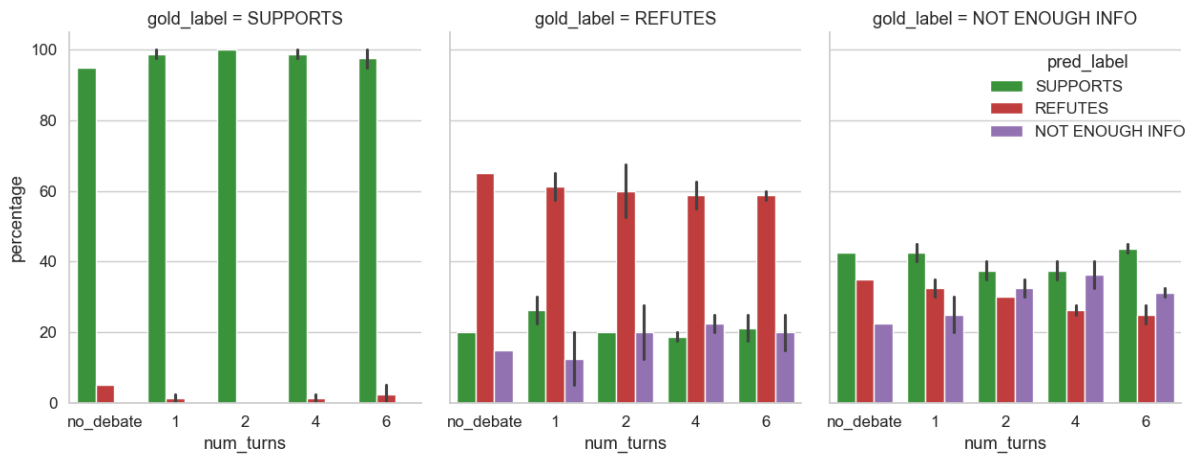


Figure 3: Prediction distribution by gold label and number of debate turns for the baseline model. Charts show percentage of predictions for each label across different debate lengths.

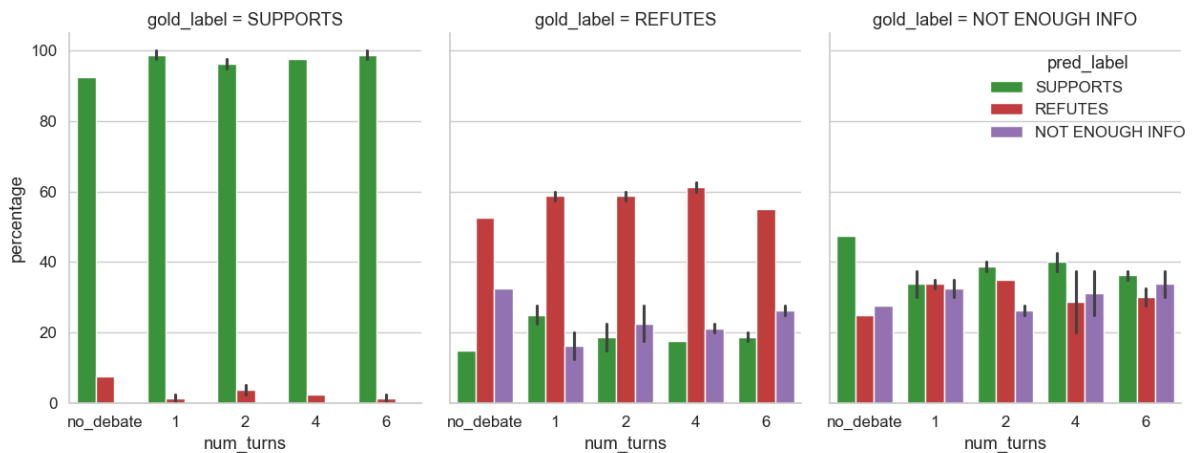


Figure 4: Prediction distribution by gold label and number of debate turns for the fine-tuned model. Charts show percentage of predictions for each label across different debate lengths

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A Appendix

A.1 Prompts Used in Experiments

Below are the exact prompt templates employed for each experimental setting.

A.1.1 No-Debate Prompt

Fact Verification Task:
Claim: {claim}
Available Evidence: {evidence}
Based on this information, classify the claim as one of the following:
- SUPPORTS
- REFUTES
- NOT ENOUGH INFO
Answer:

A.1.2 Debate Agent Prompt

Claim: "{claim}"
Evidence: "{evidence}"

You are Agent {A/B}. Your fixed role is to SUPPORT/REFUTE the claim. You must only generate arguments that support/refute the claim, regardless of how strong or weak the evidence may appear. Use the provided evidence explicitly. Do NOT hedge, switch roles, or say the evidence is inconclusive. Address specific arguments made by Agent {opponent}, and try to persuade a neutral judge. Limit your response to 3 sentences.

Debate so far:
Agent A: ...
Agent B: ...
[Reminder: Agent {agent} must SUPPORT/REFUTE the claim.]
Agent {agent}:

A.1.3 Final Judgment Prompt

Claim: "{claim}"
Evidence: "{evidence}"

Here is a debate about whether the evidence supports the claim:
Agent A: ...
Agent B: ...

Final Task: Based on the above, classify the claim as one of the following:
- SUPPORTS
- REFUTES
- NOT ENOUGH INFO
Answer:

A.2 Prompt Engineering Evolution

Prompt design played a pivotal role in shaping model behavior across both debate and no-debate settings. In early iterations, prompt formulations were often under-specified, resulting in undesirable model behaviors such as repeating the claim,

disregarding the assigned stance, or hedging responses. To address these issues, we progressively refined the prompts to guide model behavior more effectively.

In the debate setting, we introduced several key improvements:

- Clear, explicit stance instructions tied to agent roles
- Inclusion of turn-by-turn dialogue history
- Constraints against hedging or neutrality
- Direct instructions to engage with opposing arguments
- Role reminders before each agent turn

These refinements resulted in more consistent and on-topic responses. Early prompts often failed to enforce strict role alignment or argumentative engagement, leading agents to repeat prior turns or stray from their assigned position. For instance, a typical early prompt was structured as follows:

Claim: "claim" Evidence: "evidence"
Your role is to support/refute the claim using the evidence and respond thoughtfully to your opponent's arguments. Reference specific points made by Agent opponent where appropriate, and try to persuade a neutral judge. Limit your response to 3 sentences. Debate so far: Agent A: ... Agent B: ... Agent agent:

To improve adherence to the assigned stance and discourage neutrality, we introduced stronger constraints and clearer behavioral guidance:

You are Agent A/B. Your fixed role is to SUPPORT/REFUTE the claim. You must only generate arguments that support/refute the claim, regardless of the evidence's strength. Use the provided evidence explicitly. Do NOT hedge, switch roles, or say the evidence is inconclusive. Respond to arguments made by Agent opponent, and persuade a neutral judge. Limit your response to 3 sentences. Debate so far: Agent A: ... Agent B: ... [Reminder: Agent agent must SUPPORT/REFUTE the claim.] Agent agent:

These adjustments substantially improved response quality. Agents were more likely to maintain stance consistency, refer to evidence, and engage in direct rebuttals. Nonetheless, occasional lapses in role alignment persisted, suggesting that some level of inconsistency is inherent to current language models.

In the no-debate (classification) setting, prompt clarity similarly influenced performance. Our initial prompt was:

Fact Verification Task: Claim: claim
Available Evidence: evidence Based on this information, the claim is: (SUPPORTS/REFUTES/NOT ENOUGH INFO)
Answer:

This minimal prompt often yielded the default label *NOT ENOUGH INFO*, reducing classification accuracy to around 20

Fact Verification Task: Claim: claim
Available Evidence: evidence Based on this information, classify the claim as one of the following: - SUPPORTS - REFUTES - NOT ENOUGH INFO Answer:

This reformulation alone improved accuracy to approximately 60

A.3 Fine-Tuning Overfitting Examples

Early fine-tuning experiments produced responses that often resembled casual Reddit-style conversation rather than structured, evidence-based fact verification. For example, given the claim:

"Gabrielle Union was in a movie"

the model replied:

"I saw the movie. The Birth of a Nation was playing at the end of the year. Not sure what's the date, but I'd think it would be around the end of 2016."

Likewise, in response to:

"Cosmos: A Spacetime Odyssey secured studio support thanks to Seth MacFarlane"

the model generated:

"You're comparing a show to a movie, which is the same thing. A movie has a plot and a theme... this is the series to see. There's a good chance it'll change your opinion."

These outputs are loosely related to the topic but fail to directly engage with the claim or the provided evidence, and do not perform the intended verification task. We assume this behavior reflects overfitting to the informal conversational tone of the Reddit-style training data, causing the model to prioritize fluency and relatability over task alignment. After reducing the fine-tuning schedule from 5 to 2 epochs, model responses became more concise, claim-focused, and evidence-grounded—suggesting improved generalization and reduced stylistic drift.